

Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization experienced a Denial of Service (DoS) attack caused by an ICMP flood that compromised the internal network for two hours. A misconfigured firewall allowed external actors to overwhelm network infrastructure with excessive echo requests, rendering internal services unreachable. The incident management team contained the attack by blocking inbound ICMP traffic, offlining non-essential services, and restoring critical workloads. Long-term hardening measures, including rate limiting, source IP verification, network monitoring, and IDS/IPS filters, were deployed.
Identify	• **Attack Type:** ICMP Flood (Denial of Service - DoS). • **Root Cause:** Misconfigured edge firewall permitting unbounded inbound ICMP traffic. • **Affected Assets:** Internal network bandwidth, routing infrastructure, web development, and digital marketing operational servers. • **Business Impact:** Operational downtime of two hours, disrupting employees' access to internal network resources and client deliverables.

Protect	• **Firewall Hardening: Implemented strict inbound ICMP rate limiting rules to avoid bandwidth exhaustion.** • **Anti-Spoofing: Enforced source IP address verification at the perimeter firewall to identify and drop spoofed packets.** • **Network Segmentation: Isolated internal production workloads from external-facing access points.** • **Policy & Audit: Established mandatory, recurring security reviews of all perimeter firewall and router access control lists (ACLs).**
Detect	• **Network Monitoring:** Deployed network performance and continuous traffic monitoring tools to baseline throughput and detect abnormal spikes in real time. • **IDS/IPS Deployment:** Integrated signature-based and anomaly-based Intrusion Detection/Prevention Systems to inspect, flag, and drop malicious ICMP payloads. • **Log Auditing:** Centralized firewall and routing logs into an analysis pipeline to detect reconnaissance scanning patterns before service degradation occurs.
Respond	• **Containment:** Immediately severed inbound ICMP flood traffic at the firewall perimeter level. • **Traffic Mitigation:** Placed non-essential internal network services offline to reserve bandwidth and computing capacity for mission-critical operations. • **Triage & Forensics:** Analyzed packet captures to isolate attack vectors, identify attack patterns, and determine malicious source IP blocks.

	• **Communication:** Disseminated clear incident status briefings to IT staff, internal stakeholders, and client-facing teams.
Recover	• **System Restoration:** Brought critical network services back online incrementally while monitoring CPU, memory, and link saturation. • **Post-Incident Validation:** Verified integrity across all internal resources to confirm no secondary compromises or lateral movements occurred during the DoS window. • **Process Improvement:** Updated disaster recovery and incident response playbooks (IRP) with precise runbooks for volumetric network attacks.

Reflections/Notes: Applying the NIST Cybersecurity Framework highlighted that a simple perimeter misconfiguration can completely compromise service availability. Through this analysis, I learned the importance of adopting a defense-in-depth strategy: perimeter hardening (firewall rate limiting and anti-spoofing) must be backed by continuous network monitoring and IDS/IPS to catch anomalies early. Additionally, having a structured incident response and recovery plan significantly reduces downtime and ensures systematic, secure system restoration. Moving forward, regular configuration audits and automated traffic alerting will be essential to maintaining network resilience.